

CollabBoard — AI Cost Analysis

Development Period: Feb 1 – Feb 21, 2026

LLM API (Claude) Total Spend: \$41.07

- Input Token Cost: \$26.21

- Output Token Cost: \$14.86

Primary Model: Claude Sonnet 4.6

Firestore Infrastructure Usage (All within free tier):

- Realtime Database: 6.2 MB bandwidth, 379 B storage

- Cloud Functions: 66 invocations

- Hosting: 1.7 MB downloads, 8.6 MB storage

- Authentication: 39 active users

- Firestore: 4.1K writes, 29K reads, 1.9K deletes (stress test peaks briefly exceeded daily free quota)

Other AI Costs: \$0 (No embeddings, no vector database)

Total Development AI & Infrastructure Cost: \$41.07

Production Cost Assumptions:

1. 5 AI commands per session: AI is assistive (template creation, layout, edits) rather than continuously invoked.

2. 8 sessions per user per month: Approx. 2 sessions per week for collaborative use cases.

3. 40 commands per user per month: 5 commands × 8 sessions.

4. 1,600 tokens per command: Estimated average (1,200 input / 400 output) based on stress-test patterns, reduced to reflect production context pruning and fewer debugging prompts.

5. \$0.003 per command: Derived from estimated 1,600 tokens at Claude pricing; conservative blended estimate.

Monthly Cost Projections (Linear Scaling Model):

- 100 users: ~\$12/month

- 1,000 users: ~\$120/month

- 10,000 users: ~\$1,200/month

- 100,000 users: ~\$12,000/month

Model Characteristics: Costs scale linearly with command volume. No fine-tuning, embeddings, or vector storage costs assumed. Increased command frequency or token size increases cost proportionally.